Decodable, Not Cashable

A Preprint

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registered 2026-08-10 | run ledger #1–#2, #5–#6 | 4 seeds, 12/12 strict inequalities | replicated before publication | promoted 2026-08-11

Abstract

We built a reservoir designed for tidy, addressed storage — a card index — and raced it against the lab's standard mixing reservoir on the same text, same readouts, same everything. The card index stores history **strictly deeper** (its past can be decoded further back) and predicts the next character **strictly worse**, in all four seeds tested, at both spectral radii, 12/12 strict inequalities on replication. The gap grows with spectral radius: depth +0.9 → +3.0 characters, prediction cost +0.07 → +0.15 bits. A follow-up proves the failure is structural: handing the readout explicit linear "unbinding" keys to every drawer is a readout reparametrization and cannot help even in principle — the gap survives (0.081 → 0.091 at ρ=0.95). A crossover sweep locates where the trade inverts: below ρ* ≈ 0.72 the tidy delay line actually cashes better. Storing memories and using them are different talents, demonstrated on purpose.

Keywords reservoir computing · echo state networks · permutation matrices · holographic reduced representations · memory vs. prediction · pre-registration

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§1 The question

The lab's standard machine is a "pond": N = 1024 fixed random tanh neurons, sparse random recurrence, reading text one character at a time — a reservoir computer in the classical echo-state sense [1][2][3]. It never trains; only a small readout on top does. A pond *mixes* — every step smears the incoming character into a shared, entangled state.

What if it filed instead? Replace the mixing matrix with $W = \rho \cdot P$, where P is a single random 1024-cycle permutation: the state no longer mixes, it *rotates*, each step moving the whole memory one drawer along the cycle. Give the inputs Hadamard codes — 27 exactly orthogonal input columns (scaled $1/\sqrt{3}$), one per character — so deposits don't collide: the permutation-and-superposition storage of holographic reduced representations [4] and hyperdimensional computing [5], built into the recurrence. That is the **addressed pond**: a filing cabinet where yesterday's character sits j drawers along, undisturbed. Intuition says organized storage should help. Does it?

Two rulers, and the whole finding lives in their disagreement:

- **Decode depth** (the U3 ruler): how many characters back a ridge probe can still read out of the state — the interpolated 50%-accuracy crossing over lags 1–32.
- **Cashing**: validation bits-per-character (bpc) of the house logistic readout predicting the *next* character. Lower is better. For scale on these splits: uniform guessing 4.7549, 3-gram 2.9279, 5-gram 2.1585.

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§2 The machine

One registered 2×2(×2) factorial, 24 cells: recurrence W ∈ {`perm`: ρ·P; `sparse`: fanin-10 random, rescaled to spectral radius ρ} × input coding Win ∈ {`had`: Hadamard; `rnd`: uniform random} × activation {tanh, linear}, with tanh at ρ ∈ {0.6, 0.95, 1.1}. Data: 2M training / 500k validation / 500k test characters of text8 [6], washout 200. `perm_had` is the addressed pond; `sparse_rnd` is the standard control. One useful physical fact: a permutation is norm-preserving, so the addressed pond runs stably at ρ > 1, past the edge where mixing ponds normally collapse.

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§3 What we registered in advance

Six ordinal predictions and four quantitative bands, committed before the run (wording finalized after a disclosed 100k-character smoke test — a disclosure the registration protocol requires). The headline bet, O6: the addressed pond would be *both* strictly deeper on decode *and* strictly worse on logistic bpc than the standard pond at ρ ∈ {0.95, 1.1} — the claim that storage is bought by giving up mixing, stated as a conjunction so that either half failing would break it.

Because O6 was a surprising claim, the house rules made it merely provisional until a registered replication on seeds 1–3: 6 cell-pairs, 12 strict inequalities, with the falsifiers named in writing before the rerun — *any* depth inequality breaking would demote the claim to an open question; any bpc inequality breaking (depth intact) would half-falsify it; both breaking would file seed 0 as a candidate outlier. 12/12 was the only outcome allowed to promote.

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§4 What happened

Run #1 (seed 0): 6/6 ordinal predictions hit, 4/4 quantitative bands hit. The registered headline, in numbers:

Table 1. Addressed pond (`perm_had`, tanh) vs standard pond (`sparse_rnd`, tanh), seed 0. Depth in characters (higher = stores deeper); logistic val bpc (lower = predicts better).

ρ	depth perm_had	depth sparse_rnd	bpc perm_had	bpc sparse_rnd
0.95	11.05	10.11	2.7538	2.6727
1.1	13.72	10.63	2.8779	2.7337

Deeper on the left, worse on the right, at both radii — and the W-axis depth gap *grows* with ρ (roughly +0.4 → +1.0 → +3.2 characters across ρ = 0.6 / 0.95 / 1.1), with no edge-of-chaos collapse on the addressed arm. The replication, seeds 1–3:

Table 2. Registered replication (run ledger #2), seeds 1–3. Every registered inequality strict: 12/12.

seed	ρ	depth: perm > sparse	bpc: perm > sparse
1	0.95	11.04 > 10.10	2.7613 > 2.6681
1	1.1	13.70 > 10.64	2.8833 > 2.7303
2	0.95	11.02 > 10.12	2.7480 > 2.6751
2	1.1	13.68 > 10.70	2.8699 > 2.7360
3	0.95	11.03 > 10.07	2.7565 > 2.6671
3	1.1	13.67 > 10.63	2.8744 > 2.7209

The seed-to-seed spread is unexpectedly small: across four seeds the addressed pond's depth spans 11.02–11.05 at ρ=0.95 and 13.67–13.72 at 1.1 — a spread of ≤ 0.05 characters, about half a percent of the gap being tested. Plausibly because a random N-cycle is the same object up to relabeling, the depth of an addressed pond is essentially deterministic in the seed.

Can the depth be cashed with a key? A natural objection: perhaps the readout simply cannot *reach* the drawers. We therefore handed it the keys explicitly — features $[x \parallel P^{-1}x \parallel \dots \parallel P^{-4}x]$, the state unbound, in the holographic-representation sense [4], by the pond's own cycle. The registered analysis proved this vacuous before the run: each appended block is a fixed coordinate permutation of x, so the widened features span *exactly* the raw state's linear function class — a linear readout already owns every fixed permutation of its input. The run confirmed the theorem's observable edge: ridge validation identical to four decimals in every arm, and the perm–sparse logistic gap *survived*, 0.0811 → 0.0906 at ρ=0.95 and 0.1443 → 0.1484 at ρ=1.1. Linear unbinding cannot recover the stored depth even in principle. (A nonlinear drawer-opener is a named, not-yet-run follow-up.)

Where the trade inverts. A registered low-ρ sweep (seed 0, random input coding both arms, isolating the W axis) measured the gap ladder, gap(ρ) = sparse − perm logistic val bpc (positive = the delay line wins):

Table 3. The mixing-tax crossover (run ledger #6, seed 0, single-seed measured context).

ρ	perm_rnd bpc	sparse_rnd bpc	gap	winner
0.3	2.8969	2.9028	+0.0059	delay line
0.45	2.7758	2.7865	+0.0107	delay line
0.6	2.6843	2.6989	+0.0146	delay line
ρ* = 0.7227	**–**	**–**	**0**	**crossover**
0.8	2.6611	2.6519	−0.0092	mixing
0.95	2.7353	2.6727	−0.0626	mixing

Below $\rho^* \approx 0.72$ the pure delay line cashes *better*; above it, mixing wins and the tax grows. Storage ordering — the addressed pond decodes deeper — held at every ρ tested, 0.3 through 1.1. Both architectures have their bpc optimum near ρ=0.8, just above the crossover: at the operating point where these ponds predict best, filing and mixing are nearly equivalent, and they only differentiate away from it — mixing degrades gracefully, the delay line collapses.

§5 What it means

Storing memories neatly and using them are different talents, and here the trade is built on purpose: the permutation buys addressable, collision-free storage by giving up nonlinear mixing, and the readout pays for the missing mixing in bits. The wall is structural, not a missing tool — even the key to every drawer doesn't help, because "apply a fixed permutation" is something a linear reader could always do for itself. What the prediction readout wants is not deep, tidy, decodable storage; it wants the computed, entangled summaries that mixing provides. Decodable ≠ cashable.

The one honest escape hatch left open: only *linear* unbinding is ruled out. Whether any cheap *nonlinear* reader can cash the stored depth is a named open question in the lab's queue. Until then, the drawers stay shut.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. W. Maass, T. Natschläger, H. Markram, "Real-time computing without stable states: a new framework for neural computation based on perturbations," *Neural Computation* 14(11), 2002.
3. M. Lukoševičius, H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review* 3(3), 2009.
4. T. A. Plate, "Holographic reduced representations," *IEEE Transactions on Neural Networks* 6(3), 1995.
5. P. Kanerva, "Hyperdimensional computing: an introduction to computing in distributed representation with high-dimensional random vectors," *Cognitive Computation* 1(2), 2009.
6. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **169b7c2d** — `2026-08-10-addressed-pond`: main factorial, 24 cells, seed 0 (run ledger #1). Registered before running; all six ordinals and four bands hit.
- **866d975e** — `2026-08-11-addressed-seeds-replication`: seeds 1–3, 12 cells (run ledger #2). 12/12 strict inequalities; promotion by the pre-stated rule.
- **472d4fb7** — `2026-08-11-addressed-unbinding-readout`: the linear key-ring test, seed 0 (run ledger #5). Null registered in advance and confirmed; axis closed.
- **4052cdec** — `2026-08-11-addressed-low-rho-crossover`: the crossover sweep, seed 0 (run ledger #6). Single-seed measured context, labeled as such.

All experiments registered before running, with named falsifiers. Headline claim tested on 4 seeds (0–3); replicated before publication. Unbinding and crossover results are single-seed and presented as supporting structure, not as the promoted claim.